AUTO LEVELING CONTROL MODULE

id091800011000

Purpose

- The auto leveling control module calculates the target optical axis based on the vehicle height signal from the auto leveling sensor and outputs the target optical axis signal to the headlight leveling actuator.

Function

- The auto leveling control module controls the headlight auto leveling system and detects a malfunction in the headlight (LED type) circuit.

Vehicle height record function while vehicle is unoccupied

- When the auto leveling control module performs the headlight auto leveling system initial setting, it inputs the vehicle height from the auto leveling sensor and records the vehicle height while the vehicle is unoccupied.
- The vehicle height in the unloaded condition recorded by the auto leveling control module is not cleared even if the battery is disconnected.

Automatic configuration function

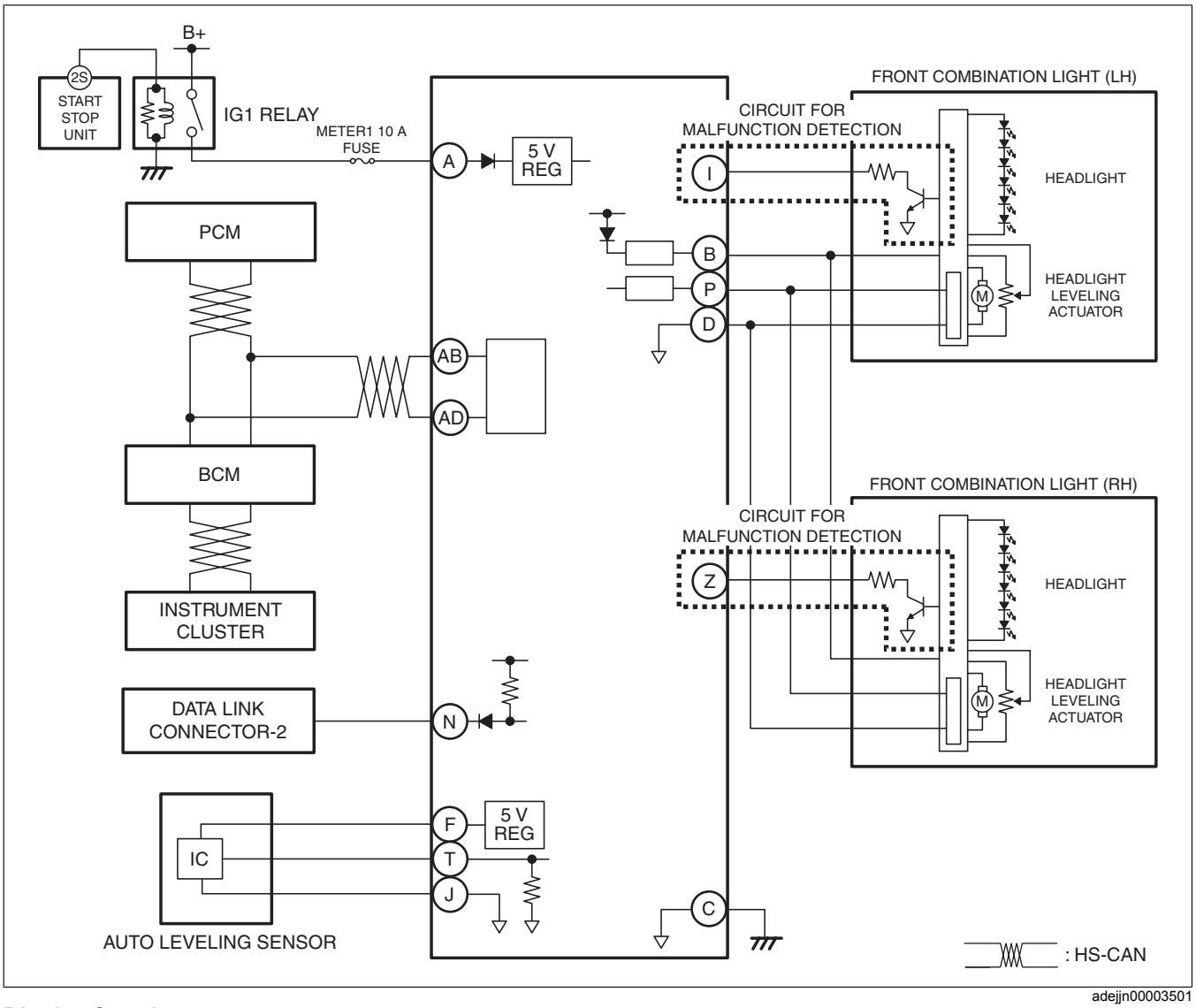
- When the ignition is switched ON (engine off or on) after the auto leveling control module is replaced, the auto leveling control module reads the vehicle specification information sent via CAN communication from the instrument cluster, and stores the vehicle specification information.

On-board diagnostic function

- The auto leveling control module is equipped with an on-board diagnosis function which records DTCs in the event of a malfunction.

Malfunction detection function

- The auto leveling control module detects a malfunction of the headlight auto leveling system related parts.
- The auto leveling control module detects a malfunction in the headlight (LED type) circuit based on the circuit for the malfunction detection equipped in the front combination light.



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Display function

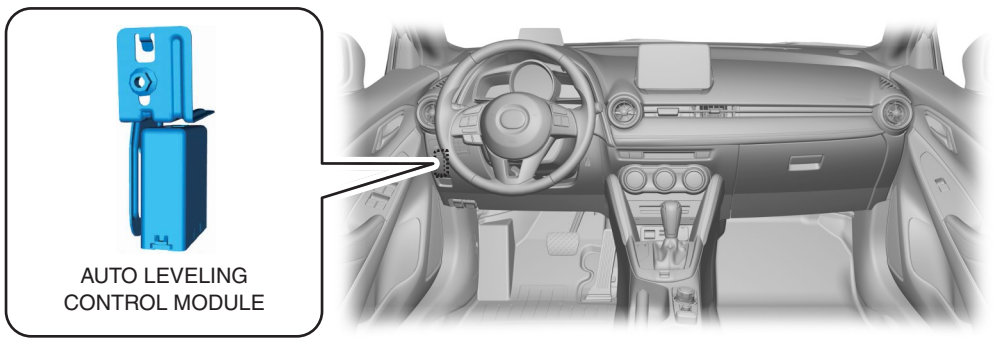
- If any malfunction is detected, the LED headlight warning light in the instrument cluster flashes to notify the driver of a system malfunction.

Burnt out bulb check function

- When the ignition is switched ON (engine off or on), the auto leveling control module turns the LED headlight warning light on for **approx. 3 s** and performs a burnt out bulb check.

Construction

- The auto leveling control module is located inside the dashboard.



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Operation

Headlight auto leveling system

- For the headlight auto leveling system operation, refer to the [HEADLIGHT AUTO LEVELING SYSTEM]. (See HEADLIGHT AUTO LEVELING SYSTEM.)

Headlight (LED type) circuit malfunction detection

- For the operation of the headlight (LED type) circuit malfunction detection, refer to the [LED HEADLIGHT WARNING LIGHT]. (See LED HEADLIGHT WARNING LIGHT.)

Fail-safe

DTC	Fail-safe
B1041:54	• The headlight leveling actuator is stopped at the position when the malfunction is determined.
B1044:01	• The headlight leveling actuator is stopped at the position when the malfunction is determined.
B1D00:13	• Control of the headlight auto leveling system is continued.
B1D01:13	• Control of the headlight auto leveling system is continued.
U0001:88	• The headlight leveling actuator is stopped at the position when the malfunction is determined.
U0100:00	• The headlight leveling actuator is stopped at the position when the malfunction is determined.
U0140:00	• The headlight leveling actuator is stopped at the position when the malfunction is determined.
U0155:00	• Control of the headlight auto leveling system is continued.
U0423:68	• Control of the headlight auto leveling system is continued.
U2005:86	• The headlight leveling actuator is stopped at the position when the malfunction is determined.
U2300:54	• Control of the headlight auto leveling system is continued.
U2300:55	• The headlight leveling actuator is stopped at the position when the malfunction is determined.
U2300:56	• Control of the headlight auto leveling system is continued.
U3000:42	• The headlight leveling actuator is stopped at the position when the malfunction is determined.
U3000:49	• The headlight leveling actuator is stopped at the position when the malfunction is determined.
U3003:16	• The headlight leveling actuator is stopped at the position when the malfunction is determined.
U3003:17	• The headlight leveling actuator is stopped at the position when the malfunction is determined.